

1 Data generation machinery for N-of-1 clinical trial simulations
2 under a two-component response decomposition

3 pmsimstats team

4 September 09, 2026

5 **Contents**

6	1 Abstract	2
7	2 1. Introduction	3
8	3 2. Methods	4
9	3.1 2.1 Notation and analysis model	4
10	3.2 2.2 Two data-generating architectures	4
11	3.3 2.3 The carryover recursion	5
12	3.4 2.4 Simulation design	6
13	4 3. Results	7
14	4.1 3.1 Power under carryover	7
15	4.2 3.2 The reference cell is ceiling-bound	7
16	4.3 3.3 Type I error	8
17	4.4 3.4 Comparison with the three-component setting	9
18	5 4. Discussion	9
19	5.1 4.1 What the TV component is for	9
20	5.2 4.2 When the reduction is appropriate	9
21	5.3 4.3 A gap this paper exposes	10
22	5.4 4.4 Limitations	10
23	6 5. Conclusions	11
24	7 Conflict of interest	11
25	8 Funding	11
26	9 Reproducibility	12
27	10 Appendix A. Why mean and covariance moderation	12
28	10.1 A.1 The two channels exhaust the Gaussian case	12
29	10.2 A.2 What each channel asserts	13
30	10.3 A.3 Standing in the literature	13
31	10.4 A.4 Holding the analysis model fixed	13

32	11 Appendix B. Correlation structure under the two-component decomposition	14
33		
34	11.1 B.1 Variable ordering and block partition	14
35	11.2 B.2 Within-component blocks A_c	15
36	11.3 B.3 The cross-component block $C_{pb,br}$	15
37	11.4 B.4 Biomarker coupling vector b	16
38	11.5 B.5 A worked example	16
39	11.6 B.6 What the reduction removes, and what it does not	17
40	12 Appendix C. Positive definiteness and the feasible range of c_{bm}	18
41	12.1 C.1 The constraint is a Schur complement on the biomarker row	18
42	12.2 C.2 Closed form for the component block under compound symmetry	19
43	12.3 C.3 Why the component count barely matters	19
44	12.4 C.4 Numerical ceilings	20
45	12.5 C.5 Practical implications	21
46	13 References	22

47 1 Abstract

48 **Background.** Simulation studies of aggregated N-of-1 trials must choose where in the joint
49 distribution a biomarker-treatment interaction lives. A companion paper compares two such
50 choices, mean moderation and covariance moderation, under a three-component decomposi-
51 tion of treatment response into pharmacological (BR), placebo-belief (PB), and time-variant
52 natural-history (TV) components. The TV component carries no biomarker coupling in either
53 architecture, so its role in that comparison is not obvious.

54 **Methods.** We repeat the comparison under a reduced data-generating process carrying only
55 BR and PB. Dropping TV reduces the correlation matrix from $(2 + 3n)$ to $(2 + 2n)$, from
56 26×26 to 18×18 at the eight occasions used throughout this program. We cross two
57 architectures against three trial designs, two sample sizes, three biomarker-moderation levels,
58 and three carryover half-lives, holding the analysis model fixed. Appendix C derives the
59 positive-definiteness constraint on the moderation parameter in closed form and characterizes
60 what the reduction does and does not relax.

61 **Results.** The architecture contrast reported by paper 01 under three components is repro-
62 duced under two. At $c_{bm} = 0.30$ and $N = 70$, raising the carryover half-life from zero to one
63 week costs covariance moderation 0.194 of power under the Hybrid design and 0.196 under
64 OL+BDC, against 0.040 and 0.078 for mean moderation. Under the crossover design neither
65 architecture loses power to carryover, because its off-drug occasions are spaced widely enough
66 that residual exposure has already decayed at every one of them. At paper 01's reference cell
67 ($c_{bm} = 0.45$, $N = 70$) all cells lie above 0.89 and the contrast is compressed by a ceiling effect
68 to 0.032 against 0.002; the reference cell understates the architecture difference roughly sixfold
69 and should not be used to read it. Empirical Type I error is strongly design-dependent and
70 architecture-independent, at 0.061 under CO, 0.039 under Hybrid and 0.021 under OL+BDC
71 against a nominal 0.05.

72 **Conclusions.** The TV component is inert for the biomarker-treatment interaction estimand
73 under the data-generating process as implemented, and acts only as a variance term. A two-
74 component decomposition is therefore adequate for methodological work whose object is a
75 comparison among architectures, designs, or analysis specifications. The full three-component
76 decomposition remains necessary when the estimand is the attribution of response to its causal
77 sources, when a candidate biomarker may correlate with a non-pharmacological component,
78 or when absolute power is reported as a design input rather than as a comparison.

79 2 1. Introduction

80 An aggregated N-of-1 trial pools repeated within-participant on-drug and off-drug contrasts to
81 estimate a population-level biomarker-treatment interaction¹. Simulation studies of such trials
82 must specify a data-generating process, and that specification involves two choices usually
83 made implicitly. The first is where the interaction lives in the joint distribution of biomarker
84 and response. The second is how many distinct latent components the response trajectory is
85 decomposed into.

86 The companion paper 01 takes up the first choice. It compares *mean moderation*, in which the
87 biomarker shifts the mean of the pharmacological response, against *covariance moderation*, in
88 which the biomarker's correlation with that response is treatment-state dependent. It shows
89 the two are not interchangeable: they respond differently to carryover, and the difference is
90 large enough to change a design recommendation.

91 The present paper takes up the second choice. The program's standard DGP decomposes
92 response into three components: BR, the pharmacological action of the compound; PB, the
93 improvement driven by the participant's belief that treatment will work; and TV, drift in
94 the underlying condition independent of treatment. That decomposition is developed and
95 defended in paper 06, whose subject is the inferential cost of failing to decompose.

96 There is a specific reason to ask whether all three are needed for *methodological* work. The
97 biomarker couples to BR in both architectures. It couples to PB only through an optional
98 contamination parameter that is zero by default. It does not couple to TV at all: the package
99 provides no parameter for a biomarker-TV correlation, so that entry of the correlation matrix
100 is identically zero in every simulation this program has run. Paper 06's omitted-variable
101 identity then implies that TV cannot displace the biomarker-treatment slope, and paper 06's
102 Study A confirms this empirically, finding no trend in bias across its TV magnitude axis.

103 If TV cannot bias the estimand, what does it do? It contributes variance. It also contributes
104 n rows and columns to the correlation matrix, two cross-component blocks, and a set of
105 positive-definiteness constraints. Whether that cost is repaid in a study whose object is the
106 comparison between architectures is the question this paper answers, by repeating paper 01's
107 comparison without it.

108 We hold everything else fixed. The trial designs, the analysis model, the parameter calibration,
109 the biomarker-moderation levels, and the carryover half-lives are those of paper 01. Section 2
110 states the model and the two architectures under the reduced decomposition, Section 3 gives

the results, and Section 4 takes up what follows for the design of future simulation work in this program. Three appendices carry the structural material: Appendix A on why the two architectures are treated as a pair, Appendix B on the correlation matrix the reduced DGP assembles, and Appendix C on the positive-definiteness constraint the reduction was expected to relax.

3 2. Methods

3.1 2.1 Notation and analysis model

Let $i = 1, \dots, N$ index participants and $p = 1, \dots, n$ measurement occasions at cumulative weeks $w_1 < \dots < w_n$. Write B_i for the participant's baseline biomarker, BL_i for baseline symptom severity, and $S_{x,ip}$ for the observed symptom score. Under the two-component decomposition,

$$S_{x,ip} = BL_i - (BR_{ip} + PB_{ip}),$$

so that a beneficial response lowers the score. The three-component model of paper 06 adds TV_{ip} inside the parentheses; setting it to zero recovers the present model exactly.

Both components are drawn jointly across occasions from a multivariate normal whose means follow modified Gompertz trajectories,

$$mg(t; a, d, r) = a \exp(-d \exp(-rt)),$$

evaluated at a component-specific time argument: cumulative time on drug t_{od} for BR, and accumulated belief exposure t_{pb} scaled by the design's expectancy weight η for PB. The distinct time arguments are what make the components separable from design contrasts.

The analysis model is held fixed across all conditions,

$$S_{x,ip} = \beta_0 + \beta_{bm} B_i + \beta_t t_p + \beta_D D_{bc,ip} + \beta_{bm:D} B_i D_{bc,ip} + u_i + \varepsilon_{ip},$$

fitted with `nlme::lme`, a participant random intercept $u_i \sim N(0, \sigma_u^2)$, and continuous-time AR(1) residual correlation `corCAR1(form = ~ t | ptID)`. Here $D_{bc,ip}$ is the continuous exposure indicator, equal to one on drug and decaying as $\exp(-\lambda t_{sd,ip})$ off drug with $\lambda = \ln 2/t_{1/2}$. The estimand is $\beta_{bm:D}$. Appendix A.4 states why the analysis model is held fixed.

3.2 2.2 Two data-generating architectures

3.2.1 2.2.1 Mean moderation (Architecture A)

Under mean moderation the biomarker shifts the conditional mean of the pharmacological component at on-drug occasions. After the multivariate normal draw, BR is displaced by

$$BR_{ip} \leftarrow BR_{ip} + c_{bm} z_i \sigma_{BR} \quad \text{for occasion } p \text{ on drug,}$$

where $z_i = (B_i - \mu_{bm})/\sigma_{bm}$ is the standardized biomarker. The covariance structure is untouched: the biomarker is independent of every component draw, and the interaction is carried entirely by the first moment. In the notation of Appendix B, mean moderation sets the coupling vector $b \equiv 0$.

3.2.2 2.2.2 Covariance moderation (Architecture B)

Under covariance moderation the means are untouched and the interaction is carried by a treatment-state-dependent correlation,

$$\text{Cor}(B_i, BR_{ip}) = c_{bm} \cdot \phi(t_{sd,ip}), \quad \phi(t_{sd}) = \begin{cases} 1, & \text{on drug} \\ \exp(-\lambda t_{sd}), & \text{off drug.} \end{cases}$$

The biomarker is more strongly correlated with the pharmacological response while the drug is present, and that association decays after discontinuation at the pharmacokinetic rate.

3.2.3 2.2.3 What the reduced decomposition changes

Neither architecture references TV. Removing it therefore leaves both definitions above unchanged and alters the simulation in three mechanical ways.

First, the correlation matrix shrinks, from $2 + 3n$ to $2 + 2n$. Appendix B gives the resulting block partition.

Second, the outcome loses a variance term. At the reference calibration the three components carry standard deviations of 10 (TV), 10 (PB), and 8 (BR), so TV accounts for approximately 38% of the component variance. Removing it reduces the residual variation against which the interaction is tested, and absolute power rises accordingly. This is an expected arithmetic consequence rather than a finding, and it is why absolute power in this paper is not comparable to paper 01's.

Third, the cross-component structure simplifies. The three-component matrix carries three cross-component blocks; the two-component matrix carries one. Appendix B.6 sets out exactly what is removed, and Appendix C shows that what is removed is not what constrains the moderation parameter.

3.3 2.3 The carryover recursion

Both architectures apply carryover to the BR mean trajectory: at an off-drug occasion, a decayed fraction of the pre-discontinuation response persists. The implementation used here differs from the one behind paper 01's published results, and the difference is worth stating.

The earlier implementation computed

172

$$\mu_{BR,p} \leftarrow \mu_{BR,p} + \mu_{BR,p-1} (1/2)^{t_{sd,p}/t_{1/2}},$$

173 in which $\mu_{BR,p-1}$ is the *already adjusted* value from the preceding occasion while $t_{sd,p}$ is
 174 *cumulative* time since discontinuation. Elapsed time is therefore counted twice, and every off-
 175 drug occasion after the first in an uninterrupted run is over-decayed. At a one-week half-life
 176 the second off-drug occasion receives half the residual effect it should.

177 The present implementation anchors the decay to the mean at discontinuation,

178

$$\mu_{BR,p} \leftarrow \mu_{BR,p} + \mu_{BR,p^*} (1/2)^{t_{sd,p}/t_{1/2}},$$

179 where p^* indexes the most recent on-drug occasion, so the cumulative decay factor is applied
 180 once. We verified at three reference cells (Hybrid, OL+BDC, and CO at $N = 70$, $c_{bm} = 0.45$,
 181 $t_{1/2} = 1.0$, 400 replicates under a common seed) that the correction moves power by at most
 182 0.005 against a Monte Carlo standard error of about 0.025. It is adopted because it is correct,
 183 not because it changes results.

184 3.4 2.4 Simulation design

185 The grid crosses architecture, trial design, sample size, biomarker-moderation strength, and
 186 carryover half-life.

Factor	Levels
Architecture	mean moderation (A), covariance moderation (B)
Trial design	CO, Hybrid, OL+BDC
Sample size N	35, 70
Biomarker moderation c_{bm}	0, 0.30, 0.45
Carryover half-life $t_{1/2}$	0, 0.5, 1.0 weeks
Components	BR + PB
Replicates	500

187 This gives $2 \times 3 \times 2 \times 3 \times 3 = 108$ cells. The $c_{bm} = 0$ rows are the Type I error check against
 188 nominal 5%; the $t_{1/2} = 0$ rows give a no-carryover reference against which the cost of nonzero
 189 carryover is read off directly.

190 Throughout, N is the **total** number of participants, allocated as evenly as possible across the
 191 design's randomization paths, so a design with P paths assigns about N/P to each.

192 Absolute power is not comparable between architectures at the same nominal c_{bm} , since
 193 the parameter is not calibrated to equal power across the two channels. The interpretable
 194 comparison is the carryover profile within each architecture, and the qualitative shape of that
 195 profile between them.

Table 1: Power by architecture, trial design and carryover half-life at $c_{bm} = 0.30$, $N = 70$, 500 replicates per cell. Cost is the power lost in moving from no carryover to a one-week half-life. Absolute power is not comparable between architectures at a common nominal c_{bm} ; the interpretable quantity is the profile across half-life within each architecture.

Architecture	Design	$t_{1/2} = 0$	0.5	1.0	Cost
Covariance (B)	CO	0.746	0.786	0.768	-0.022
Covariance (B)	Hybrid	0.878	0.778	0.684	+0.194
Covariance (B)	OL+BDC	0.746	0.610	0.550	+0.196
Mean (A)	CO	0.768	0.746	0.770	-0.002
Mean (A)	Hybrid	0.842	0.836	0.802	+0.040
Mean (A)	OL+BDC	0.754	0.692	0.676	+0.078

4 3. Results

All 108 cells converged completely: the non-convergence rate is zero throughout. At 500 replicates the Monte Carlo standard error on power does not exceed 0.022, and differences smaller than about twice that should not be over-interpreted.

4.1 3.1 Power under carryover

Table 1 carries the principal result. Under the Hybrid design, moving from no carryover to a one-week half-life costs covariance moderation 0.194 of power, against 0.040 for mean moderation. Under OL+BDC the corresponding figures are 0.196 and 0.078. Both covariance-arm costs are several times the Monte Carlo standard error; both mean-arm costs are within two or three of it. This is paper 01's central finding, reproduced on a correlation matrix two-thirds the size.

The crossover design behaves differently, and consistently for both architectures: neither arm loses power to carryover under CO, and the point estimates move by less than one standard error in either direction. The reason is structural rather than statistical. CO spaces its occasions 2.5 weeks apart, so at a one-week half-life the residual exposure multiplier $\phi(t_{sd})$ has already fallen below 0.18 at the first off-drug occasion and below 0.03 at the second. There is almost no residual exposure left for carryover to contaminate. The same feature explains why CO's positive-definiteness ceiling is insensitive to $t_{1/2}$ (Appendix C.4), so the power result and the structural result agree.

4.2 3.2 The reference cell is ceiling-bound

Table 2 repeats the comparison at paper 01's reference cell, and is included as a caution rather than as a result. Every cell lies above 0.89. The Hybrid carryover cost under covariance moderation falls to 0.032, roughly a sixth of its value at $c_{bm} = 0.30$, not because the mechanism is weaker but because there is no room above the starting value for the loss to register.

We record this explicitly because the reference cell is the natural place to look for comparability with paper 01, and doing so here would understate the architecture difference by a factor of

Table 2: The same comparison at paper 01’s reference cell ($c_{bm} = 0.45$, $N = 70$). Every entry lies above 0.89, so the architecture contrast is compressed by a ceiling effect and the carryover cost is understated.

Architecture	Design	$t_{1/2} = 0$	0.5	1.0	Cost
Covariance (B)	CO	0.982	0.992	0.976	+0.006
Covariance (B)	Hybrid	0.998	0.994	0.966	+0.032
Covariance (B)	OL+BDC	0.988	0.968	0.890	+0.098
Mean (A)	CO	0.972	0.992	0.976	-0.004
Mean (A)	Hybrid	0.992	0.986	0.990	+0.002
Mean (A)	OL+BDC	0.976	0.980	0.966	+0.010

Table 3: Empirical Type I error at the null ($c_{bm} = 0$), pooled within trial design over both architectures, both sample sizes and all three carryover half-lives (500 replicates per cell). Nominal level 0.05.

Design	cells	Rejection rate	SE	z vs 0.05
CO	12	0.061	0.0031	+3.6
Hybrid	12	0.039	0.0025	-4.5
OL+BDC	12	0.021	0.0018	-15.8

about six. The informative cells are those where baseline power is well away from one. The same caution applies in reverse at $N = 35$, where the covariance arm’s OL+BDC cost reaches 0.252.

4.3 3.3 Type I error

Type I error is not uniform across designs, and the departure is large enough to require comment. Pooled within design, the rejection rate is 0.061 under CO, 0.039 under Hybrid, and 0.021 under OL+BDC, against a nominal 0.05.

Two features of this pattern matter. It is architecture-independent: the two arms agree closely within every design, which is expected, since at $c_{bm} = 0$ the two architectures generate identical data. And it is monotone in how much off-drug information the design supplies, running from anti-conservative under CO to markedly conservative under OL+BDC.

The consequence for this paper is a restriction on what may be compared. Power figures from different designs are computed under tests with materially different actual sizes, so the cross-design power comparisons in Table 1 should be read as descriptive rather than as equal-alpha comparisons. The within-design, across-half-life comparisons that carry the paper’s argument are unaffected, since the test size is held fixed along that axis.

We did not diagnose the mechanism. The companion carryover paper attributes a milder conservatism in the same analysis model to a working covariance that does not match the data-generating one, and a cluster-robust standard error removes it there. Whether the same correction would flatten the design dependence seen here is a reasonable conjecture and an

242 untested one.

243 4.4 3.4 Comparison with the three-component setting

244 Absolute power under the two-component DGP is higher than paper 01 reports under three
245 components, and the explanation is arithmetic. At the reference calibration TV carries a
246 standard deviation of 10 against 8 for BR and 10 for PB, so it contributes approximately 38%
247 of the component variance. Removing it reduces the residual variation against which a fixed
248 interaction is tested; on a variance argument alone, standard errors shrink by roughly a fifth
249 and power rises correspondingly.

250 This comparison is qualitative and should not be given a number. Paper 01's published figures
251 were produced under the earlier carryover recursion of Section 2.3 and a partly different grid,
252 so the difference between the two sets of power values confounds the component reduction
253 with at least two other changes. Separating them would require a matched three-component
254 run under the present code, which we did not perform.

255 What can be said without a matched run is that nothing in the *structure* of the comparison
256 changed. The ranking of architectures within each design, the direction of the carryover effect,
257 the insensitivity of CO, and the design ordering are all as paper 01 reports them.

258 5 4. Discussion

259 5.1 4.1 What the TV component is for

260 The results support a narrow claim and not a broad one.

261 The narrow claim is that TV is inert for the biomarker-treatment interaction estimand. It
262 carries no biomarker coupling in either architecture, the package provides no parameter with
263 which to give it one, and paper 06's omitted-variable identity shows that a component uncor-
264 related with the biomarker cannot displace the biomarker slope regardless of its magnitude.
265 The present results are consistent with that identity and extend it in one respect: paper 06
266 established it under a single architecture, and it now holds under both.

267 The broad claim, which we do not make, is that TV is dispensable. Three of its roles survive
268 intact. It is required whenever the estimand is the attribution of response to pharmacology
269 versus belief versus natural history, which is paper 06's subject. It is required whenever
270 a candidate biomarker may correlate with natural-history drift, a case discussed in Section
271 4.3. And it contributes realism to absolute power figures calibrated against a clinical reference
272 trajectory, so any paper reporting power as a design input rather than as a comparison should
273 retain it.

274 5.2 4.2 When the reduction is appropriate

275 We suggest the following division.

276 Use the two-component DGP for methodological work whose object is a comparison: between
277 DGP architectures, between analysis specifications, between trial designs, or between inference

procedures. In that setting the third component adds variance, dimension, and positive-definiteness constraints without adding anything the comparison depends on, and its absence makes the mechanism easier to see. The saving is real but modest: the correlation matrix falls from 26×26 to 18×18 , and Appendix C shows the feasible range of c_{bm} widens by only about 0.01.

Use the three-component DGP whenever a number is intended to be read as an estimate of what a real trial would achieve, or whenever the decomposition itself is the object of study.

The reduction should not be adopted for the reason that first suggests it. One might expect that deleting a third of the rows and columns would materially enlarge the feasible parameter space. Appendix C.3 shows why it does not: under compound symmetry the smallest eigenvalue of the component block is $(1 - \rho) - (c_1 - c_x)$, which contains neither the number of occasions nor the number of components. The case for the reduction is transparency and cost, not numerical headroom.

5.3 4.3 A gap this paper exposes

Paper 06 states its central identity over both non-pharmacological components, and its conclusion recommends decomposition when a candidate biomarker may correlate with PB *or* TV. The package implements a biomarker-PB coupling and no biomarker-TV counterpart. The TV half of that claim has therefore never been exercised, and could not be without new code.

This bears on the present paper directly. Our conclusion that TV is inert holds under the DGP as implemented, in which the biomarker-TV correlation is structurally zero. It would not hold under a DGP permitting such a coupling; under paper 06's own identity it would certainly not, since a biomarker correlated with natural-history drift would displace the one-component slope exactly as a PB-correlated biomarker does. The inertness of TV is thus a property of the current implementation rather than of the component, and the natural next step is to add the missing parameter and test the half of paper 06's claim that has never been run.

5.4 4.4 Limitations

The comparison inherits paper 01's restriction to a single prazosin-PTSD parameter calibration, so absolute power values should be read as illustrative and the ranking as the more portable result.

The design-dependent Type I error of Section 3.3 limits cross-design comparison, as set out there. It was not diagnosed.

The reduction removes TV rather than flattening its trajectory. The two are not equivalent, since the code sets component standard deviations independently of the Gompertz maximum: a flattened TV mean leaves the variance contribution and the covariance blocks in place. A study wanting the simpler mental picture without the change in residual variance should flatten rather than remove.

316 No matched three-component grid was run under the present code, so Section 3.4’s comparison
317 with paper 01 remains qualitative.

318 The two architectures are simulated in separate replicate loops from different seeds, since
319 the same random draw does not correspond to the same trial under different data-generating
320 processes. Comparisons between architectures are therefore unpaired, and in any case not
321 interpretable on an absolute scale at a common nominal c_{bm} .

322 6 5. Conclusions

- 323 1. **The architecture comparison survives the reduction.** Under a BR + PB data-
324 generating process, covariance moderation continues to lose power as carryover grows
325 while mean moderation does not. At $c_{bm} = 0.30$ and $N = 70$ the cost is 0.194 against
326 0.040 under Hybrid, and 0.196 against 0.078 under OL+BDC.
- 327 2. **TV is a variance term, not a signal term.** It carries no biomarker coupling in
328 either architecture, and the package provides no means of giving it one. Its removal
329 raises absolute power without disturbing the ranking, the direction of the carryover
330 effect, or the ordering among designs.
- 331 3. **The reference cell should not be used to read the contrast.** At $c_{bm} = 0.45$ and
332 $N = 70$ every cell exceeds 0.89 and the architecture difference is compressed roughly
333 sixfold by a ceiling effect. Comparability with paper 01’s presentation is not a sufficient
334 reason to report it as the headline.
- 335 4. **Type I error is design-dependent.** Pooled rejection rates run from 0.061 under
336 CO to 0.021 under OL+BDC against a nominal 0.05, with the two architectures agree-
337 ing closely within each design. Cross-design power comparisons are correspondingly
338 descriptive rather than equal-alpha.
- 339 5. **The reduction is appropriate for comparative methodological work and not**
340 **for design estimates**, and it should be adopted for transparency and cost rather than
341 for numerical headroom, which Appendix C shows it does not supply.
- 342 6. **The inertness of TV is a property of the implementation.** It follows from the
343 absence of a biomarker-TV coupling parameter, not from anything intrinsic to natural-
344 history drift. Adding such a parameter would be the natural way to test the untested
345 half of paper 06’s central claim.

346 7 Conflict of interest

347 The authors declare no conflicts of interest.

348 8 Funding

349 This work received no specific funding.

9 Reproducibility

The simulation driver is `analysis/scripts/two-component-dgp/01-run-grid.R` and its output is versioned at `analysis/data/13-two-component-grid.rds`. The driver sets a master seed and derives a per-cell seed from it. Every number quoted in the results is read from that file at render time rather than typed in.

The two-component decomposition is supported by the `components` argument added to `generateData()` and `buildSigma()` in `R/generateData.R`. It defaults to `c("tv", "pb", "br")`, so existing calls are unaffected; passing `c("pb", "br")` produces the reduced matrix used here. The argument requires that `"br"` be retained, since BR carries the interaction channel in both architectures, and canonicalizes the ordering so that label and standard-deviation sequence is stable regardless of how the caller writes it. The package test suite passes unchanged under the default.

The corrected carryover recursion of Section 2.3 is applied in `R/generateData.R` and `implementations/tidyverse/R/functions.R`. Other implementation collections retain the earlier form; see `02-deviations.md` in the driver directory.

10 Appendix A. Why mean and covariance moderation

10.1 A.1 The two channels exhaust the Gaussian case

The simulations draw participant trajectories from a multivariate normal distribution, and a multivariate normal is completely determined by its mean vector μ and its covariance matrix Σ . Any biomarker-treatment interaction written into such a data-generating process must therefore enter through μ , through Σ , or through both. Mean moderation is the first route, covariance moderation the second, and the combined architecture developed in a companion paper² is the third. Within the Gaussian family there is no fourth place to put the interaction. This is why the two architectures are treated as a pair rather than as two entries in an open-ended list: they are not two options among many, but the two components of a decomposition.

Reducing the response decomposition from three components to two does not disturb this argument. The exhaustiveness claim concerns where an interaction can be written into a Gaussian joint distribution, not how many latent components that distribution is built from. Dropping TV removes rows from Σ ; it does not create a third place for the interaction to live, nor remove either of the two that exist.

The exhaustiveness claim is specific to the Gaussian family and should not be read more broadly. A data-generating process built on a finite mixture, or on explicit latent responder classes, can carry an interaction in features that neither μ nor Σ captures for a single component, and Senn notes finite mixtures as an alternative variance-component model³. Covariance

385 moderation is best understood as the second-moment reduced form of such a mixture, which
386 is why the latent-class treatment is developed separately⁴.

387 10.2 A.2 What each channel asserts

388 The two channels are not merely different parameterizations of one mechanism; each commits
389 to a distinct claim about what the biomarker is.

390 Mean moderation asserts that the biomarker fixes a participant's drug effect. Two participants
391 sharing a biomarker value share an expected treatment effect exactly, and any difference
392 between their realized responses is measurement error. This is the appropriate commitment
393 when the biomarker measures pathway engagement directly, or stands in a causal path between
394 treatment and response.

395 Covariance moderation asserts that the biomarker predicts response only probabilistically.
396 Two participants sharing a biomarker value may still respond differently, because the
397 biomarker indexes an unobserved responder status imperfectly rather than determining
398 response. This is the appropriate commitment when the biomarker is a statistical proxy,
399 which is the situation the prazosin and PTSD application was designed around.

400 The distinction is substantive rather than technical. It determines what a fitted interaction
401 coefficient means, and how much power carryover costs.

402 10.3 A.3 Standing in the literature

403 Mean moderation is the prevailing convention wherever biomarker-treatment interactions are
404 simulated. Haller and Ulm⁵ place the interaction in a regression mean, as do the oncology
405 design literatures on enrichment and predictive-marker trials^{6,7}. Grenet and colleagues⁸ use
406 the same form for a crossover. That literature is, however, overwhelmingly parallel-group.

407 The N-of-1 methodological literature does not adopt mean moderation, because it does not
408 model an observed moderator at all. Its standard formulation carries a random participant-
409 by-treatment interaction, a variance component describing unexplained heterogeneity with
410 no covariate attached^{3,9,10}. Biomarker-validation simulation for N-of-1 designs is therefore
411 close to unoccupied territory rather than a settled field with a convention to follow, and
412 Hendrickson et al.¹ is its principal entry. The two-channel decomposition of A.1 is adopted
413 precisely because no established N-of-1 convention exists to inherit.

414 10.4 A.4 Holding the analysis model fixed

415 Both architectures are analyzed with the identical model of Section 2.1. This is deliberate
416 and worth stating explicitly, because a data-generating process and an analysis model are
417 separate objects, and a comparison that varies both attributes differences to neither. The
418 continuous exposure variable Dbc loses contrast variance as carryover grows, and a binary on-
419 drug indicator does not. A comparison in which one architecture carried a continuous regressor
420 and the other a binary one would confound that effect with the architecture contrast it was

meant to isolate. Holding the analysis model fixed means every difference reported in Section 3 is attributable to the data-generating process alone.

The same argument governs the reduction studied here. The analysis model does not reference the component decomposition at all: it sees only S_x , and S_x is a sum. Removing TV therefore changes what the analysis is fitted to without changing what it fits, which is the property that makes the comparison in Section 3 interpretable.

427

11 Appendix B. Correlation structure under the two-component decomposition

This appendix sets out the correlation matrix the reduced data-generating process assembles, in the notation of paper 01's Appendix B so that the two can be read side by side. Symbols follow the variable names used in the code: BM is the biomarker, written B_i in Section 2.1, and BR_p , PB_p the response components at occasion p .

Mean moderation appears below only through the statement that its coupling vector is zero. It writes no biomarker-response correlation into the covariance at all; its interaction is carried in the mean instead, as Section 2.2.1 sets out.

11.1 B.1 Variable ordering and block partition

Each participant's trajectory is drawn from a single multivariate normal over an ordered variable list. With n measurement occasions at cumulative weeks $w_1 < \dots < w_n$, the two-component vector is

$$X = (\text{BM}, \text{BL}, PB_1, \dots, PB_n, BR_1, \dots, BR_n)^\top, \quad \dim X = 2 + 2n,$$

against $2 + 3n$ under the three-component decomposition. The ordering is component-major: all n occasions of one component are contiguous, rather than all components at one occasion. Writing A_c for the $n \times n$ within-component block of component $c \in \{pb, br\}$ and $C_{pb,br}$ for the single cross-component block, the correlation matrix partitions as

$$R = \begin{pmatrix} 1 & 0 & 0^\top & b^\top \\ 0 & 1 & 0^\top & 0^\top \\ 0 & 0 & A_{pb} & C_{pb,br} \\ b & 0 & C_{br,pb} & A_{br} \end{pmatrix}.$$

Three features carry over unchanged from the three-component case. The baseline term BL is uncorrelated with every other variable, so its row and column are those of the identity. The

biomarker BM couples only to the response component BR , through the vector $b \in \mathbb{R}^n$; its correlation with PB is zero throughout this paper. And b is the only place in R where the architectures differ.

The third feature deserves emphasis because it is what licenses the reduction. In the three-component matrix the biomarker's row is $(1, 0, 0^\top, 0^\top, b^\top)$: zero against both TV and PB . Deleting the TV rows and columns therefore deletes only zeros from that row. The coupling vector is untouched, entry for entry.

11.2 B.2 Within-component blocks A_c

Let ρ_c denote the autocorrelation parameter of component c (`c.pb`, `c.br` in the code). The correlation decays with the elapsed time between occasions, using the cumulative week gap rather than the index difference:

$$[A_c]_{ij} = \rho_c^{|w_i - w_j|}.$$

For $n = 4$, writing $d_{ij} = |w_i - w_j|$,

$$A_c = \begin{pmatrix} 1 & \rho_c^{d_{12}} & \rho_c^{d_{13}} & \rho_c^{d_{14}} \\ \rho_c^{d_{12}} & 1 & \rho_c^{d_{23}} & \rho_c^{d_{24}} \\ \rho_c^{d_{13}} & \rho_c^{d_{23}} & 1 & \rho_c^{d_{34}} \\ \rho_c^{d_{14}} & \rho_c^{d_{24}} & \rho_c^{d_{34}} & 1 \end{pmatrix}.$$

Because the exponent uses calendar time, unequally spaced designs produce unequal decay across consecutive occasions. In the Hybrid design, whose occasions fall at weeks 4, 8, 9, 10, 11, 12, 16, 20, consecutive gaps of one week during the discontinuation phase yield much higher adjacent correlations than the four-week gaps at either end.

We note that the code uses the cumulative week gap, not the occasion index difference, and that this choice is consequential: Appendix C.4 reports that the index-based form would cost roughly a fifth of the feasible range of c_{bm} .

11.3 B.3 The cross-component block $C_{pb,br}$

Let c_1 denote the same-occasion cross-component correlation (`c.cf1t`) and $c_\times$ the different-occasion cross-component correlation (`c.cfct`). The same-occasion entry is a constant and the different-occasion entries inherit the AR(1) decay:

$$[C_{pb,br}]_{ij} = \begin{cases} c_1, & i = j \\ c_\times \rho^{|w_i - w_j|}, & i \neq j. \end{cases}$$

Under the three-component decomposition there are three such blocks, $C_{tv,pb}$, $C_{tv,br}$ and $C_{pb,br}$, all of this form. The reduction retains the last and deletes the first two.

477 11.4 B.4 Biomarker coupling vector b

478 The vector b carries the biomarker-treatment interaction under covariance moderation. Its
 479 entry is gated on the drug indicator and decays with time since discontinuation $t_{sd,p}$ at rate
 480 $\lambda = \ln 2/t_{1/2}$:

$$481 \quad b_p = \begin{cases} c_{bm}, & \text{occasion } p \text{ on drug} \\ c_{bm} e^{-\lambda t_{sd,p}}, & \text{occasion } p \text{ off drug.} \end{cases}$$

482 Under mean moderation $b \equiv 0$ and the interaction is carried by the displacement of μ_{BR}
 483 described in Section 2.2.1. The two architectures are therefore distinguished, in the language
 484 of this appendix, by exactly one object: whether b is populated or zero.

485 11.5 B.5 A worked example

486 To make the blocks concrete, take four occasions drawn from the Hybrid design at cumulative
 487 weeks $w = (10, 11, 12, 16)$, with the first and fourth on drug and the second and third off it, so
 488 that $t_{sd} = (0, 1, 2, 0)$. Set $\rho_{br} = 0.6$, $\rho_{pb} = 0.4$, $c_1 = 0.3$, $c_\times = 0.1$, $c_{bm} = 0.45$, and a carryover
 489 half-life of one week, giving $\lambda = \ln 2 \approx 0.693$. The matrix of week gaps is

$$490 \quad (|w_i - w_j|) = \begin{pmatrix} 0 & 1 & 2 & 6 \\ 1 & 0 & 1 & 5 \\ 2 & 1 & 0 & 4 \\ 6 & 5 & 4 & 0 \end{pmatrix}.$$

491 The unequal spacing matters: occasions one to three are a week apart, while the fourth sits
 492 four weeks after the third.

493 11.5.1 B.5.1 Within-component block A_{br}

494 With $\rho_{br} = 0.6$,

$$495 \quad A_{br} = \begin{pmatrix} 1 & 0.6000 & 0.3600 & 0.0467 \\ 0.6000 & 1 & 0.6000 & 0.0778 \\ 0.3600 & 0.6000 & 1 & 0.1296 \\ 0.0467 & 0.0778 & 0.1296 & 1 \end{pmatrix}.$$

496 The fourth occasion is nearly uncorrelated with the first three, a direct consequence of the
 497 four-week gap: $0.6^4 = 0.1296$ against $0.6^1 = 0.6$ for an adjacent pair.

498 11.5.2 B.5.2 Cross-component block $C_{pb,br}$

499 With $c_1 = 0.3$, $c_\times = 0.1$ and the decay taken at $\rho = 0.6$,

$$C_{pb,br} = \begin{pmatrix} 0.3000 & 0.0600 & 0.0360 & 0.0047 \\ 0.0600 & 0.3000 & 0.0600 & 0.0078 \\ 0.0360 & 0.0600 & 0.3000 & 0.0130 \\ 0.0047 & 0.0078 & 0.0130 & 0.3000 \end{pmatrix}.$$

The block is symmetric here because the two components share a decay rate in this example; in general it need not be.

11.5.3 B.5.3 Biomarker vector b

Occasions one and four are on drug, occasions two and three off it with $t_{sd} = 1$ and 2. With $c_{bm} = 0.45$ and $\lambda = \ln 2$,

$$b = (0.4500, 0.2250, 0.1125, 0.4500)^\top.$$

The off-drug entries halve with each additional week since discontinuation, and the fourth occasion returns to full strength because the participant is back on drug. It is this alternation that carries the interaction, and its erosion under carryover that Section 3 measures.

11.6 B.6 What the reduction removes, and what it does not

Relative to the three-component matrix, the reduction removes:

- one within-component block, A_{tv} ($n \times n$);
- two cross-component blocks, $C_{tv,pb}$ and $C_{tv,br}$;
- n rows and columns from R , taking it from $2 + 3n$ to $2 + 2n$: at $n = 8$, from 26×26 to 18×18 .

It removes no entry of b , because the biomarker never coupled to TV . It removes no parameter from the architecture definitions of Section 2.2, because neither references TV .

One distinction is worth recording, because it is easy to conflate two operations that are not equivalent. Removing the component is not the same as flattening its trajectory. The code sets component standard deviations independently of the Gompertz maximum, so setting that maximum to zero yields a flat TV mean while leaving the TV variance contribution, its block A_{tv} , and its cross-component blocks in place. A study wanting the simpler mental picture without the change in residual variance should flatten; a study wanting the smaller matrix should remove. This paper removes.

12 Appendix C. Positive definiteness and the feasible range of

c_{bm}

The correlation matrix of Appendix B must be positive definite for the multivariate normal draw to be valid. It is not automatically so: the moderation parameter c_{bm} is bounded above, and the bound depends on the correlation structure, the trial design, and the number of occasions. This appendix derives the bound, gives a closed form for the compound-symmetric case, and reports where the reduction of Appendix B does and does not relax it.

The motivation is practical. Simulation drivers in this program repair a non-positive-definite matrix silently, by projection onto the nearest positive-definite matrix, so a grid cell that violates the bound does not fail. It quietly becomes a different data-generating process, with a realized biomarker-response correlation no longer equal to the nominal c_{bm} . Knowing the bound in advance is therefore a precondition for trusting a grid.

12.1 C.1 The constraint is a Schur complement on the biomarker row

Partition the correlation matrix of Appendix B.1 by separating the biomarker from everything else. Because BL is uncorrelated with all other variables, it contributes an identity block that factors out of every determinant and may be dropped. What remains is

$$R' = \begin{pmatrix} 1 & \tilde{b}^\top \\ \tilde{b} & M \end{pmatrix}, \quad M = \begin{pmatrix} A_{pb} & C_{pb,br} \\ C_{br,pb} & A_{br} \end{pmatrix},$$

where M is the $Kn \times Kn$ component block ($K = 2$ here, $K = 3$ under the full decomposition) and $\tilde{b} \in \mathbb{R}^{Kn}$ is the biomarker coupling extended by zeros: it equals b on the BR positions and vanishes elsewhere.

By the Schur complement, R' is positive definite if and only if M is positive definite and

$$1 - \tilde{b}^\top M^{-1} \tilde{b} > 0.$$

Writing $\tilde{b} = c_{bm} v$, where v is the unit-scaled coupling pattern with $v_p = \phi(t_{sd,p})$ on the BR positions and zero elsewhere, the condition becomes $c_{bm}^2 v^\top M^{-1} v < 1$, giving the exact ceiling

$$c_{bm}^* = (v^\top M^{-1} v)^{-1/2}. \tag{1}$$

Three consequences follow immediately, and they organize the rest of this appendix. The ceiling depends on the component block only through the quadratic form $v^\top M^{-1} v$, so components that are weakly coupled to BR contribute little to it. It depends on the design through v , which encodes the on-drug and off-drug pattern. And it is unaffected by mean moderation, under which $b \equiv 0$ and the constraint reduces to positive definiteness of M alone.

556 12.2 C.2 Closed form for the component block under compound symmetry

557 When the within-component blocks are compound symmetric rather than AR(1), M admits an
 558 exact spectral decomposition. This is the structure of the published implementation discussed
 559 in paper 01, and it is worth recording because it explains a result of C.3 that is otherwise
 560 surprising.

561 Under compound symmetry with a common within-component correlation ρ , common same-
 562 occasion cross-component correlation c_1 , and common different-occasion cross-component cor-
 563 relation $c_\times$, the component block is a Kronecker sum,

$$564 \quad M = A \otimes I_n + B \otimes (J_n - I_n), \quad (2)$$

565 where J_n is the $n \times n$ matrix of ones and A, B are $K \times K$:

$$566 \quad A = I_K + c_1(J_K - I_K), \quad B = \rho I_K + c_\times(J_K - I_K).$$

567 A describes the same-occasion structure across components, B the across-occasion structure.
 568 Since $J_n - I_n$ has eigenvalues $n - 1$ once and -1 with multiplicity $n - 1$, and I_n commutes
 569 with it, the Kn eigenvalues of M collapse to those of two $K \times K$ reductions:

$$570 \quad \text{eig}(M) = \underbrace{\text{eig}(A + (n - 1)B)}_{\text{once each}} \cup \underbrace{\text{eig}(A - B)}_{\text{multiplicity } n-1}.$$

571 Both reductions are of the form $\alpha I_K + \beta(J_K - I_K)$, whose eigenvalues are $\alpha + (K - 1)\beta$ once
 572 and $\alpha - \beta$ with multiplicity $K - 1$. Substituting gives all four distinct eigenvalues in closed
 573 form:

$$\begin{aligned} &\text{all-ones time direction: } [1 + (n - 1)\rho] + (K - 1)[c_1 + (n - 1)c_\times], \\ &\quad [1 + (n - 1)\rho] - [c_1 + (n - 1)c_\times] \quad (K - 1 \text{ times}), \\ 574 &\text{time-contrast directions: } (1 - \rho) + (K - 1)(c_1 - c_\times), \\ &\quad (1 - \rho) - (c_1 - c_\times) \quad (K - 1 \text{ times, each with multiplicity } n - 1). \end{aligned}$$

575 At the program's reference values $\rho = 0.7$, $c_1 = 0.2$, $c_\times = 0.1$, $n = 8$, the three-component
 576 block has exactly four distinct eigenvalues, 7.7, 5.0, 0.5 and 0.2, the last with multiplicity
 577 $2(n - 1) = 14$.

578 12.3 C.3 Why the component count barely matters

579 The smallest eigenvalue in C.2 is generally the last,

580

$$\lambda_{\min}(M) = (1 - \rho) - (c_1 - c_{\times}), \quad (3)$$

581 and the striking feature of (3) is what does not appear in it. It contains neither n nor K .
 582 Under compound symmetry the positive-definiteness of the component block is independent
 583 of both the number of occasions and the number of components. Adding a third component
 584 does not make M harder to keep positive definite; nor does adding occasions.

585 This is the analytic explanation for a result that would otherwise be puzzling. One might ex-
 586 pect that deleting a third of the rows and columns from a correlation matrix would materially
 587 enlarge the feasible parameter space. It does not, and (3) shows why: the binding structural
 588 constraint never depended on the count in the first place.

589 Two further readings of (3) are worth recording. The operative cross-component quantity
 590 is the *difference* $c_1 - c_{\times}$, not either coefficient separately, so keeping the same-occasion and
 591 across-occasion couplings close to one another matters more than keeping either small. And
 592 positive definiteness requires $\rho < 1 - (c_1 - c_{\times})$, which at the reference values is $\rho < 0.9$: a
 593 compound-symmetric structure cannot represent very high autocorrelation at all.

594 Under AR(1) the corresponding statement is different, and no equally compact closed form
 595 is available, because A_c is Toeplitz rather than exchangeable and does not commute with the
 596 cross-component structure in the way (2) requires. Numerically, the AR(1) component block
 597 becomes progressively more ill-conditioned as occasions are added, since each new occasion
 598 is strongly correlated with its neighbours. The number of *occasions* therefore does constrain
 599 c_{bm} under AR(1), while the number of *components* still does not.

600 12.4 C.4 Numerical ceilings

601 Evaluating (1) directly at the program's reference parameters ($\rho = 0.7$, $c_1 = 0.2$, $c_{\times} = 0.1$,
 602 $n = 8$, AR(1) in cumulative weeks), minimized across each design's randomization paths:

Design	$t_{1/2}$	Three components	Two components
CO	0.0	0.619	0.627
CO	0.5	0.619	0.627
CO	1.0	0.619	0.627
Hybrid	0.0	0.456	0.469
Hybrid	0.5	0.506	0.516
Hybrid	1.0	0.544	0.552
OL+BDC	0.0	0.452	0.464
OL+BDC	0.5	0.500	0.509
OL+BDC	1.0	0.533	0.540

603 The reduction buys between 0.008 and 0.013 of feasible range. This is consistent with C.3: the
 604 deleted blocks were coupled to *BR* only through c_1 and $c_{\times}$, both small, so they contributed
 605 little to the quadratic form $v^{\top} M^{-1} v$ that sets the ceiling.

606 Three further observations follow from evaluating (1) under variations of the structure, none
607 of which is specific to the reduction.

608 First, the ceiling rises with the carryover half-life within the two non-crossover designs, from
609 0.456 to 0.544 under Hybrid. A longer half-life makes the coupling vector v smoother, and a
610 smoother v aligns better with the well-conditioned directions of M . The crossover design is
611 insensitive to $t_{1/2}$ because its off-drug occasions are spaced at least 2.5 weeks apart, so $\phi(t_{sd})$
612 has decayed to near zero at every one of them regardless.

613 Second, the number of occasions matters considerably more than the number of components.
614 Holding the design shape fixed and varying n alone, the ceiling falls from approximately 0.81
615 at $n = 4$ to 0.58 at $n = 12$ under a three-component structure. That is an order of magnitude
616 larger than the effect of deleting an entire component.

617 Third, the exponent scale in A_c is consequential. Using the occasion-index difference $|i -$
618 $j|$ in place of the cumulative week gap $|w_i - w_j|$, an otherwise identical AR(1) structure,
619 lowers the worst-cell ceiling from approximately 0.45 to 0.35. The code uses the week gap;
620 documentation elsewhere in this project has described it as the index difference, and the two
621 are not interchangeable.

622 12.5 C.5 Practical implications

623 For the present paper, (1) confirms that the grid of Section 2.4 is feasible throughout: the
624 largest moderation level used is $c_{bm} = 0.45$, against a worst-cell ceiling of 0.464 under the two-
625 component structure. The margin is real but not generous, and a study wishing to extend the
626 grid past $c_{bm} = 0.45$ should evaluate (1) first rather than rely on the simulation to complain,
627 since it will not.

628 More generally, the appendix identifies which levers move the feasible range and which do not.
629 The number of latent components does not, and neither does the choice between two and three
630 of them. The number of measurement occasions does, substantially. The within-component
631 correlation ρ does, non-monotonically, with the reference value 0.7 close to the maximum of
632 the ceiling. And the cross-component coefficients matter through their difference rather than
633 their magnitudes.

634 **12.5.0.1 Provenance.** The Kronecker identity (2) and the eigenvalue decomposition of
635 C.2 were verified numerically against direct construction for $K \in \{2, 3\}$ and $n \in \{4, 8, 12\}$.
636 The ceilings in C.4 were computed from (1) and agree with an independent grid search over
637 c_{bm} in steps of 0.01 to within the grid step. The observations in C.4 concerning occasion
638 count, exponent scale, and ρ were obtained by the same grid search under a reimplemented
639 matrix builder rather than the production `buildSigma`, and should be read as characterizing
640 the structure rather than as production audit results.

13 References

1. Hendrickson RC, Thomas RG, Schork NJ, Raskind MA. Optimizing aggregated N-of-1 trial designs for predictive biomarker validation: Statistical methods and theoretical findings. *Frontiers in Digital Health*. 2020;2:13. doi:10.3389/fdgth.2020.00013
2. pmsimstats team. A combined data-generating architecture for biomarker-treatment interactions: Channel super-additivity and mixing-weight sensitivity in aggregated N-of-1 trials. Published online 2026.
3. Senn S. Sample size considerations for n-of-1 trials. *Statistical Methods in Medical Research*. 2019;28(2):372-383. doi:10.1177/0962280217726801
4. pmsimstats team. Latent-class and mixture-model formulations for biomarker-treatment interaction in N-of-1 trials. Published online 2026.
5. Haller B, Ulm K. A simulation study on estimating biomarker-treatment interaction effects in randomized trials with prognostic variables. *Trials*. 2018;19(1):128.
6. Simon R. Clinical trial designs for evaluating the medical utility of prognostic and predictive biomarkers in oncology. *Personalized Medicine*. 2010;7(1):33-47.
7. Freidlin B, Korn EL. Biomarker enrichment strategies: Matching trial design to biomarker credentials. *Nature Reviews Clinical Oncology*. 2014;11(2):81-90.
8. Grenet G, Blanc C, Bardel C, Gueyffier F, Roy P. Comparison of crossover and parallel-group designs for the identification of a binary predictive biomarker of the treatment effect. *Basic and Clinical Pharmacology and Toxicology*. 2020;126(1):59-64. doi:10.1111/bcpt.13293
9. Senn S. Mastering variation: Variance components and personalised medicine. *Statistics in Medicine*. 2016;35(7):966-977. doi:10.1002/sim.6739
10. Wang Y, Schork NJ. Power and design issues in crossover-based N-of-1 clinical trials with fixed data collection periods. *Healthcare (Basel)*. 2019;7(3):84.